

TECHNICAL SKILLS

AI Agents & LLMs	LLMs, Multi-Agent Systems, Tool Calling (MCP), RAG, Prompt Grounding, Context & Memory Graphs, Fine-Tuning
Voice & Multimodal	LiveKit (WebRTC), Silero VAD, Streaming STT/TTS (Deepgram, ElevenLabs), Turn Detection, TensorFlow, PyTorch
Software Engineering	Python (FastAPI, asyncio), TypeScript, JavaScript (Node.js, React, Vite), Kotlin (Android Compose), C/C++ (FreeRTOS), REST APIs
Hardware & Systems	3D CAD Modeling (STL), Wearable Packaging, ESP32-S3 Firmware (I2S DMA, Opus, OTA), Kubernetes Helm, Git, Linux

EXPERIENCE

Founder, Aura	Jun 2025 – Present
Wearable AI · Embedded Systems · Edge Integration · Open Source	
Pune, India	
- Built an ESP32-S3 wearable audio pendant: modeled custom 3D enclosures and wrote FreeRTOS firmware for 16 kHz I2S DMA audio capture, Opus compression, and OTA updates (<code>firmware/src/mic.cpp</code>). - Integrated device with open-source Omi ecosystem via a FastAPI backend for real-time streaming, transcription, and memory storage. - Diagnosed Windows Electron GPU memory leak from CSS backdrop filters over WebGL (authored Issue #8438, merged downstream in PR #8902). - Authored offline speaker clustering using SciPy cosine linkage with 52 unit tests to solve speaker drift in meetings (PR #8919, adopted downstream in PR #12471).	
AI & ML Intern, AICTE via Google	Oct 2024 – Dec 2024
Computer Vision · TensorFlow · ML Kit · Mobile Edge Deployment	
Pune, India	
- Trained and benchmarked lightweight on-device vision models with TensorFlow and Google ML Kit for mobile edge classification under memory limits. - Applied transfer learning and data augmentation pipelines to optimize model inference latency and accuracy on mobile hardware.	
Freelance Developer, AI & Automation	Sep 2024 – Present
Workflow Automation · LLM Tool Calling · Rapid Prototyping	
Remote	
Built custom AI agent workflows and automation scripts for developer projects, integrating LLM tool calling, structured outputs, and automated test gates.	

PROJECTS

Mia: Personal AI Engineering OS	github.com/thesohamdatta/Mia
TypeScript · Bun · Multi-Host LLM Adapters · Vitest · JSONL State Engine	
- Built a compiled TypeScript CLI (single Bun binary) automating a 5-stage engineering workflow (grill, spec, plan, review, ship) with unified adapters for Claude, OpenAI, and local models. - Engineered immutable JSONL state tracking with automated verification gates (lint, typecheck, Vitest), replacing background daemons with stateless execution (ADR-0001).	
Voice AI Systems: Real-Time Voice Agents & Mobile Clients	GitHub / Voice-AI
LiveKit WebRTC · Silero VAD · ElevenLabs · Python · Kotlin (Android Jetpack Compose)	
- Built real-time conversational voice agents on LiveKit WebRTC, implementing client-side Silero VAD for sub-second barge-in interruption and turn detection. - Developed John, a native Android client (Kotlin, Jetpack Compose) with tool calling, and voicecoder, a VS Code extension for voice-driven code editing with multi-model fallback.	
LLM-Council: Multi-Model Consensus Engine	github.com/thesohamdatta/LLM-Council
Python · FastAPI · React · Vite · Async Multi-Provider Orchestration	
- Built a full-stack consensus engine querying OpenAI, Anthropic, Gemini, and Groq concurrently to cross-evaluate and synthesize model responses. - Implemented a 3-stage critique pipeline (Analyst, Skeptic, Synthesizer) with an interactive React frontend showing real-time model agreement.	

EDUCATION

University of Pune (SPPU)	Pune, India
Bachelor of Engineering (B.E.) in Artificial Intelligence & Machine Learning	
2022 – Jun 2026	
Core Coursework: Computer Vision, Edge AI Inference, LLMs & Multi-Agent Systems, Deep Learning, Distributed Systems, Algorithms	